Foundations of Artificial Intelligence

CaseHub: Where Justice Meets Intelligence

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report " **CaseHub: Where Justice Meets Intelligence** " is the bonafide work of "**Haritha.M** , **Valluru Varshini**" who carried out the project work under my supervision.

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ABSTRACT

Judicial inefficiency and communication gaps continue to be long-standing challenges in legal ecosystems, particularly for judges, lawyers, and clients navigating complex case workflows. To address these issues, CaseHub, a comprehensive Legal Management System, has been designed to optimize and streamline court case handling, scheduling, and communication. This platform creates real-time collaboration among judges, lawyers, and clients through a cross-platform application tailored for legal use.Judges can view and manage their schedules, access summaries of past hearings, make private annotations, and get real-time transcriptions and AI-generated summaries during court proceedings. Lawyers can upload documents, record voice notes, track hearing schedules, and get notified about legal updates instantly. Clients benefit from simplified access to case progress and can communicate with their legal representatives through a secure interface.An admin panel provides centralized control over user roles, manages legal documents, tracks activities, and ensures compliance through audit logs. The system supports document management, OTP-verified login, multi-language support (English, Hindi, Tamil), and advanced filtering for case search.CaseHub leverages a modern tech stack including Flutter for a cross-platform responsive frontend, Firebase for secure authentication and real-time data sync, Firestore for document- based storage, and AI/ML models like BERT and RoBERTa to enhance decision prediction and case summarization. It also uses Socket.IO to deliver real-time notifications and updates with low latency.

INTRODUCTION

In many courts and legal institutions, the daily experience for judges, lawyers, and clients— especially those managing multiple ongoing cases—is often marked by confusion, fragmented communication, and unpredictable delays. Hearings may be rescheduled without notice, case files may be scattered, and there's often no centralized way for legal stakeholders to stay informed or collaborate effectively. Recognizing these persistent pain points, CaseHub – The Legal Management Platform was created.CaseHub is not just another case-tracking tool—it is a thoughtfully engineered, AI-powered, real-time legal management platform that enhances communication, transparency, and efficiency within the judicial system. The goal was to build a smart solution that directly addresses everyday frustrations in legal workflows while meaningfully applying current and emerging technologies.With features like real-time hearing updates, AI-based hearing summaries, document sharing, judge-specific annotations, and role-based dashboards, CaseHub empowers each user group— judges, lawyers, and clients— with better control, accountability, and visibility. The platform was developed using a design thinking methodology, placing a strong emphasis on understanding the needs of each user segment and continuously iterating to develop features that genuinely add value to the legal process.This report outlines how design thinking was strategically integrated at every stage—from empathy and ideation to prototyping and implementation—to deliver a smart, scalable, and human-centered solution to the challenges of modern legal systems.

LITERATURE REVIEW

To support the development of CaseHub, an extensive review of related literature was conducted, spanning the domains of legal case management systems, judicial technology, and the application of design thinking in software development. This research validated key assumptions, revealed limitations in existing systems, and highlighted opportunities for innovation in the legal tech space.

Several studies focused on the inefficiencies within traditional legal workflows, particularly in courts lacking centralized digital infrastructure. A recurring theme was the lack of real-time scheduling updates, fragmented communication between stakeholders, and manual handling of legal documents—all contributing to delayed justice and administrative overload. Existing legal management tools were often case-centric but not user-centric; they lacked features like role-specific dashboards, AI-powered hearing summaries, or secure annotation capabilities, which are particularly beneficial in complex judicial ecosystems.

In the area of design thinking, various academic and industry sources stressed how this methodology bridges the gap between technological development and actual human needs. Notably, a study by Brown & Wyatt (IDEO) emphasized how iterative prototyping and constant user feedback loops lead to more satisfactory and sustainable solutions. The Stanford Design Thinking model, in particular, has been successfully applied in civic tech and public service domains—demonstrating that human-centered design can transform outdated systems into efficient, empathetic, and inclusive platforms.

Collectively, these findings provided a robust foundation for the development of CaseHub. The literature confirmed that integrating real-time functionality, role-aware system design, and continuous user involvement throughout the development process would result in a more intelligent, scalable, and reliable legal management platform. It reinforced the need for a structured approach like design thinking to ensure the final product aligns not just with technical specifications, but with the nuanced needs of judges, lawyers, and clients navigating the legal system every day.

2.1 Legal Case Document Summarization: Extractive and Abstractive Methods and Their Evaluation

Authors: Abhay Shukla, Paheli Bhattacharya

Problem Addressed: Long legal documents pose challenges for quick understanding and retrieval of essential information.

Methodology: Comparative study between extractive and abstractive summarization approaches using datasets from the Indian and UK Supreme Courts. Limitations: Restricted to case law datasets; abstractive models exhibit hallucination risks.

Key Takeaway: Extractive methods ensure factual accuracy, while abstractive methods offer conciseness but risk errors — a vital balance for judicial contexts.

Relevance to CaseHub: Informs CaseHub's hearing summarization module, suggesting a hybrid extractive- first pipeline to enhance factual reliability.

2.2 Classification of U.S. Supreme Court Cases Using BERT-Based Techniques

Authors: Shubham Vatsal, Adam Meyers

Problem Addressed: Automating the classification of Supreme Court judgments for large-scale legal organization.

Methodology: BERT-based models trained on 15 broad and 279 fine-grained categories of legal cases. Limitations: Domain limited to the U.S. Supreme Court; requires high computational resources (GPUs). Key Takeaway: Transformer-based models excel in legal classification but demand domain-specific tuning and hardware optimization.

Relevance to CaseHub: Helps in developing automated case-tagging and similar-case retrieval modules.

2.3 Incorporating Domain Knowledge for Extractive Summarization of Legal Case Documents (DELSumm)

Authors: Paheli Bhattacharya, Soham Poddar

Problem Addressed: Generic summarizers fail to capture legal-specific context and terminology.

Methodology: Domain-aware extractive summarization enhanced with legal ontologies.

Limitations: Focused solely on Indian Supreme Court data; lacks multilingual adaptability.

Key Takeaway: Domain-aware summarizers yield higher relevance without extensive supervision.
Relevance to CaseHub: Guides development of accurate judge-facing summaries using legal ontologies and contextual signals.

2.4 How Ready Are Pre-Trained Abstractive Models and LLMs for Legal Case Judgment Summarization?

Authors: Aniket Deroy, Kripabandhu Ghosh
Problem Addressed: Reliability and hallucination issues in LLM-generated summaries.
Methodology: Evaluation of GPT-3, GPT-4, and domain-specific LLMs on summarization benchmarks.
Limitations: Requires human validation; factual inaccuracies still occur.
Key Takeaway: LLMs are fluent but need verification mechanisms like RAG and human-in-the-loop evaluation.
Relevance to CaseHub: Motivates integration of RAG and factual-checking layers for summaries.

2.5 AI-Powered Legal Intelligence System Architecture (LICES)

Authors: Kalaycioglu et al., 2025
Problem Addressed: Integrating speech, text, and document analysis in unified legal systems. Methodology: Hybrid architecture combining speech recognition, NLP pipelines, and multi-layered LLMs. Limitations: Prototype stage; lacks scalability validation.
Key Takeaway: Multimodal integration (audio + text) enhances court automation but needs robust scaling. Relevance to CaseHub: Inspires CaseHub's multimodal transcription and summarization components (TTS/ STT integration)

2.6 Mining Legal Arguments in Court Decisions

Author: Ivan Habernal (2023)
Problem Addressed: Extracting structured arguments from lengthy judicial texts.
Methodology: Argument mining using expert-annotated ECHR datasets and NLP-based classification.
Limitations: Limited to European Court of Human Rights; not yet adapted for Indian judiciary.

Key Takeaway: Structured argument extraction supports legal reasoning and search efficiency.
Relevance to CaseHub: Enables features like “similar legal reasoning” retrieval and argumentative summarization.

2.7 A Multilingual Approach to Predicting Legal Judgments in the European Court of Human Rights
Authors: Chalkidis et al. (2019)
Problem Addressed: Multilingual challenges in judicial outcome prediction.
Methodology: Multilingual BERT models trained across ECHR datasets.
Limitations: Lower performance in low-resource languages.
Key Takeaway: Cross-lingual models enable multilingual predictions but struggle in underrepresented languages.
Relevance to CaseHub: Reinforces the need for multilingual NLP (English, Hindi, Tamil) for broader accessibility.

2.8 Automatic Classification of Legal Cases: Predicting the Top Court’s Rulings
Authors: Sulea et al. (2017)
Problem Addressed: Automation of judicial decision prediction.
Methodology: SVM and NLP-based classification of French Supreme Court cases.
Limitations: Domain-limited; language-specific.
Key Takeaway: Traditional ML techniques are efficient on specific corpora but lack generalization. Relevance to CaseHub: Acts as a baseline for CaseHub’s judgment prediction modules using advanced transformer models.

2.9 JusticeAI: A Large Language Models Inspired Collaborative and Cross-Domain Multimodal System for Automatic Judicial Rulings in Smart Courts Author: Nagwan Abdel Samee (2024)
Problem Addressed: Enhancing judicial decision-making through hybrid ensemble
Methodology: Cross-domain ensemble using pre-trained embeddings and LLMs for smart court rulings.
Limitations: Tested only on ECHR datasets; limited jurisdictional validation.
Key Takeaway: Ensemble fusion enhances robustness but must be tuned for fairness and localization. Relevance to CaseHub: Serves as a design reference for CaseHub’s Cross-Domain Neural Knowledge Fusion (CDKF) system.

PROPOSED SYSTEM

The proposed system, CaseHub, is an AI-powered legal management and collaboration platform designed to revolutionize the way court cases are handled, monitored, and analyzed. It aims to overcome the limitations of traditional legal workflows that often involve paper-based documentation, fragmented communication, and manual scheduling. By integrating advanced technologies such as Artificial Intelligence (AI), cloud computing, and real-time synchronization, CaseHub provides a unified, intelligent, and user-friendly platform that enhances efficiency, transparency, and accessibility in the judicial system.The system offers dedicated portals for judges, lawyers, and clients, each customized according to their role and responsibilities. Judges can manage hearings, view case insights, and access AI- generated summaries; lawyers can upload case documents, receive notifications, and record voice notes; while clients can track case progress and communicate securely. These role-based dashboards ensure that every stakeholder interacts only with features relevant to them, reducing complexity and increasing productivity.To enable secure and seamless access, Firebase is used for authentication, real-time database management, and cloud storage. It ensures secure login, encrypted document handling, and instant updates across all connected devices.

The integration of Google APIs for Speech-to-Text (STT) and Text-to-Speech (TTS) allows real-time voice transcription and playback, making hearings, notes, and summaries accessible in English, Hindi, and Tamil.The AI layer of CaseHub employs techniques such as Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to generate accurate and context-aware summaries, retrieve similar past cases, and provide data-driven insights to assist in judicial decision-making. These models are fine-tuned using Hugging Face Transformers to better understand legal terminology and reasoning. Additionally, a constraint satisfaction algorithm has been implemented to automatically allocate courtrooms and schedule hearings efficiently based on availability and case priority.The system's modular design ensures flexibility and scalability for future enhancements, such as AI- powered judgment prediction, voice assistant integration, and mobile application support.

ARCHITECTURE DIAGRAM

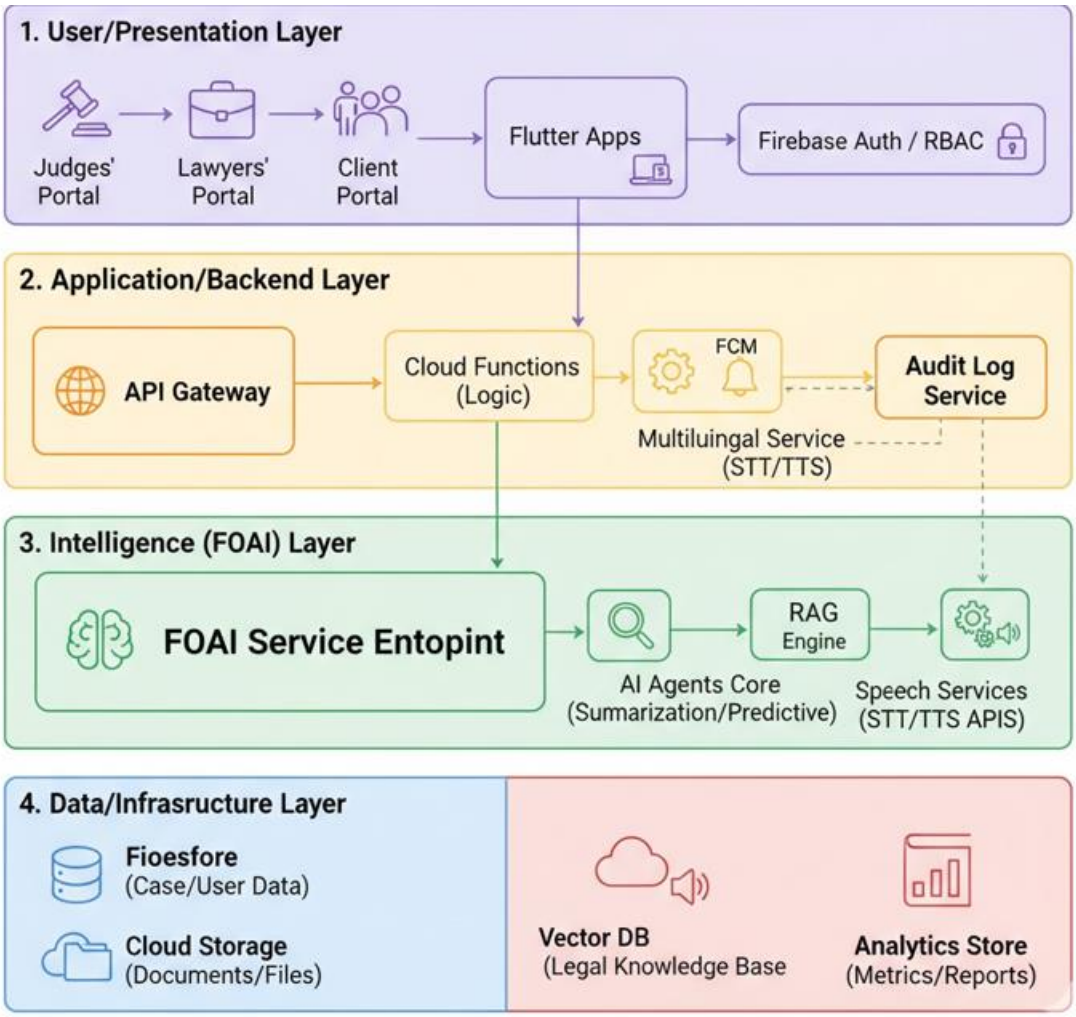

Fig 3.1.1 ARCHITECTURE DIAGRAM FOR CASE HUB

MODULES DESCRIPTION

The CaseHub system is structured into multiple interconnected modules that collectively aim to simplify case handling, enhance collaboration, and integrate artificial intelligence into judicial processes. Each module is designed to serve a specific role within the legal ecosystem—judges, lawyers, and supporting systems—while maintaining seamless communication and secure data flow between them. Through features like AI-driven summarization, real-time updates, and intelligent document management, CaseHub transforms traditional legal workflows into a smart, efficient, and data-informed process. This modular approach ensures scalability, flexibility, and adaptability for future advancements in smart court systems.

4.1 Module 1: Judge Portal

The Judge Portal provides judges with intelligent tools for managing hearings, tasks, and accessing real- time AI-generated case insights.

Key Features:

- Case scheduling & task management
- Secure notes & annotations
- AI-based hearing transcription & summaries
- Similar case insights using RAG

AI Techniques Used:

1. Speech-to-Text (STT) & Text-to-Speech (TTS) with Google Translator

Converts courtroom speech to text using Google STT, translates it through Google Translate, and replays it via Google TTS.

This voice text voice workflow allows multilingual accessibility in English, Hindi, and Tamil, making hearing transcripts and summaries easy to understand for diverse users.

2. Retrieval-Augmented Generation (RAG)

Retrieves relevant case documents and similar past judgments from a vector database and integrates them with LLMs to produce legally grounded summaries and insights.

This helps judges review accurate, context-based case summaries and precedents efficiently.

Purpose:

To assist judges in making informed, data-driven decisions through automated hearing summaries, precedent retrieval, and multilingual accessibility.

Module 2: Lawyer Portal

The Lawyer Portal is designed to help lawyers organize their cases, upload documents, and receive updates instantly through AI-enhanced functionalities.

Key Features:

•Case list & hearing schedule

•Document upload & voice notes

•Real-time updates & user registration

AI Techniques Used:

•Large Language Models (LLMs)

Transformer-based LLMs (such as BERT, RoBERTa, ALBERT, GPT) are used to interpret legal documents, generate concise summaries, and provide natural-language insights.

Fine-tuned using Hugging Face datasets, these models ensure high-quality understanding of legal content and context.

•Programming Languages & Frameworks

◦Python: Used for AI/ML model development and backend automation.

◦HTML, CSS, JavaScript: For frontend web design and interactive interfaces.

◦Constraint Satisfaction Algorithm: Used for efficient room allocation and scheduling of hearings.

Purpose:
To support lawyers in managing documentation, preparing cases effectively, and staying updated with real- time insights and scheduling powered by AI.

IMPLEMENTATION AND RESULT

The CaseHub system was implemented using a combination of modern web technologies, cloud infrastructure, and AI frameworks. The frontend was developed using Flutter for cross-platform compatibility, ensuring that judges and lawyers can access the system seamlessly through both web and mobile devices. The backend utilized Firebase for authentication, data storage, and real-time synchronization, enabling secure login, role-based access, and instant notifications. Artificial Intelligence modules were integrated using Python and Hugging Face Transformers to handle summarization, retrieval, and natural language understanding. The Speech-to-Text (STT) and Text-to-Speech (TTS) services were powered by Google APIs, allowing multilingual accessibility in English, Hindi, and Tamil. The Retrieval-Augmented Generation (RAG) mechanism and Large Language Models (LLMs) were used to generate accurate legal summaries, case insights, and argument retrievals, ensuring factual and legally grounded outputs. Additionally, a constraint satisfaction algorithm was implemented to automate the allocation of courtrooms and hearing schedules efficiently.
During testing, the system successfully performed secure user authentication, real-time case synchronization, document uploads, and AI-based summarization. Judges were able to view automatically generated hearing summaries and retrieve similar past cases, while lawyers could manage their case lists, upload files, and receive instant updates.

The results demonstrated that CaseHub significantly improved case management efficiency, reduced manual documentation time, and enhanced decision-making with reliable AI-generated insights. The multilingual support and real-time updates further contributed to a smoother judicial workflow, validating the system's effectiveness as an AI-powered smart court management platform.

5.1 EXPERIMENTAL SETUP

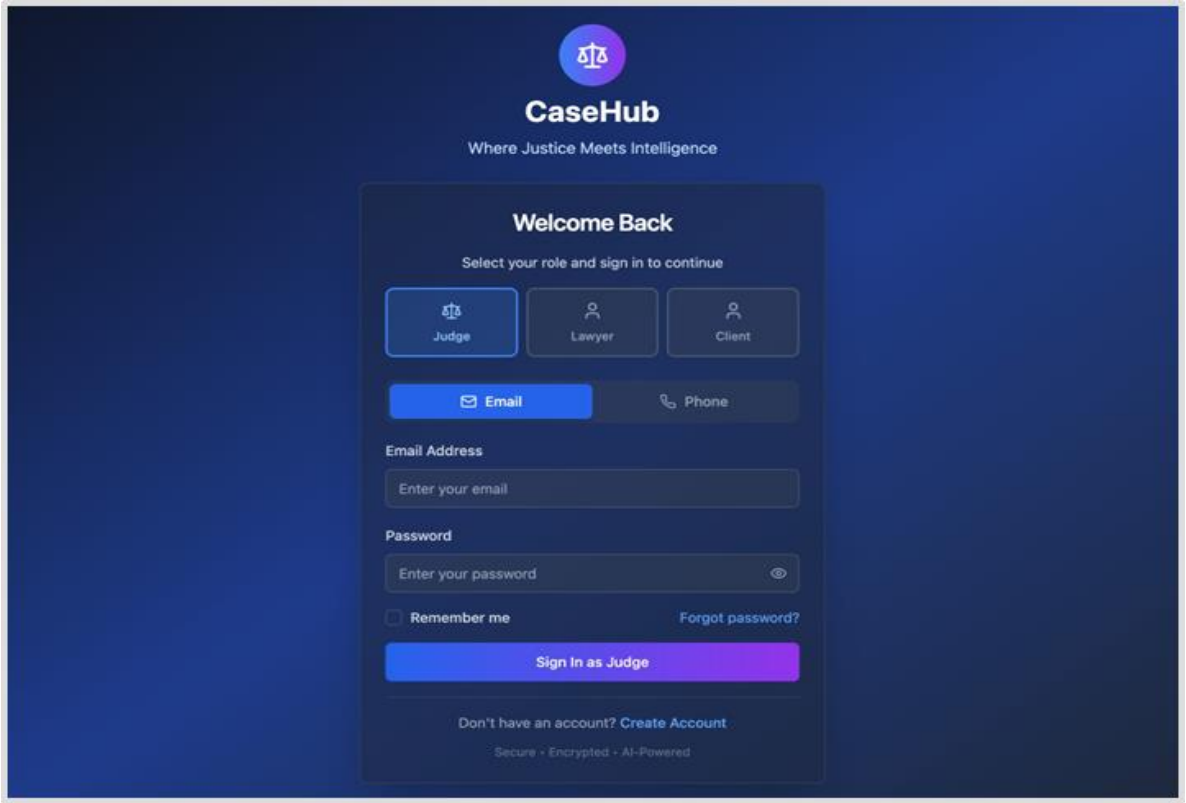

Fig 5.1.1 CASE HUB WELCOME PAGE

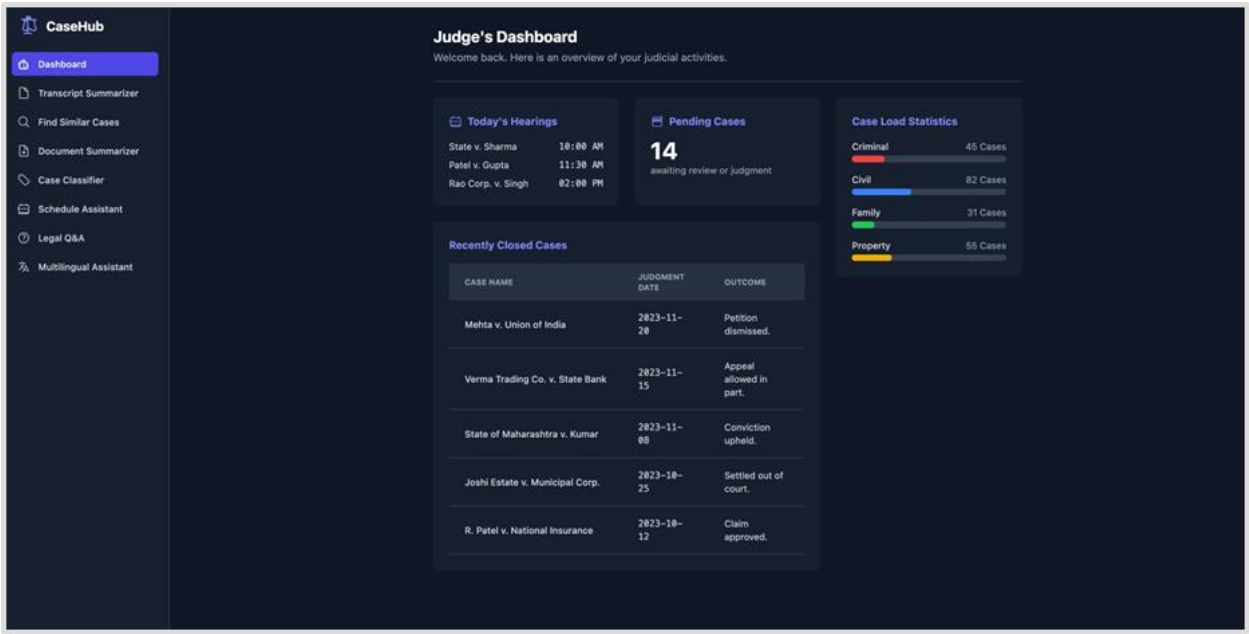

Fig 5.1.2 CASE HUB JUDGE DASBOARD

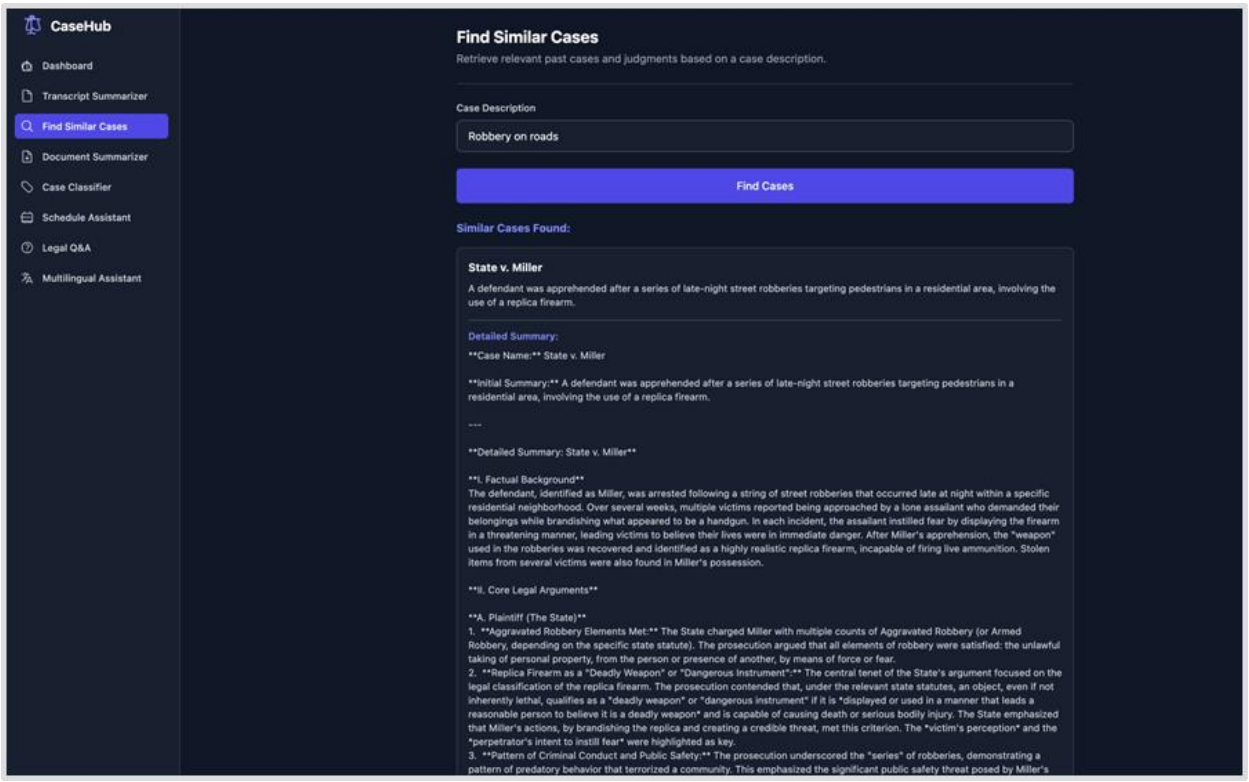

Fig 5.1.3 CASE HUB SIMILAR CASES DASHBOARD

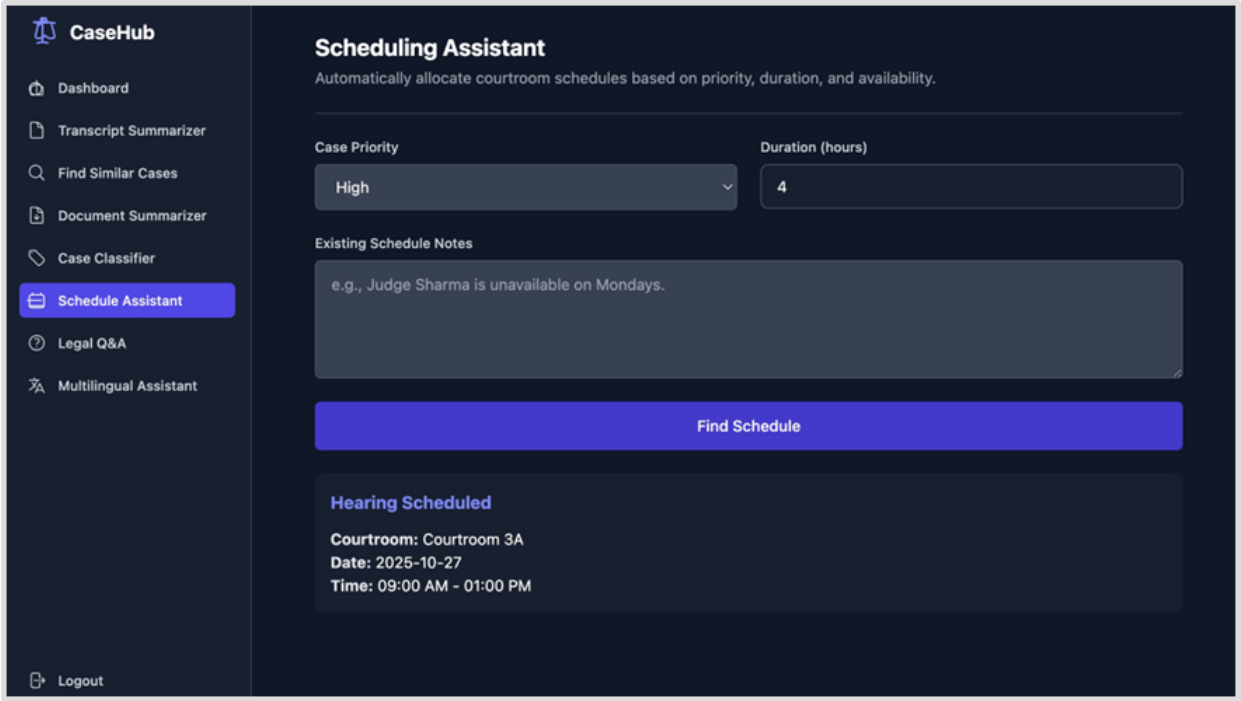

Fig 5.1.4 CASE HUB SCHEDULE ASSISTANT

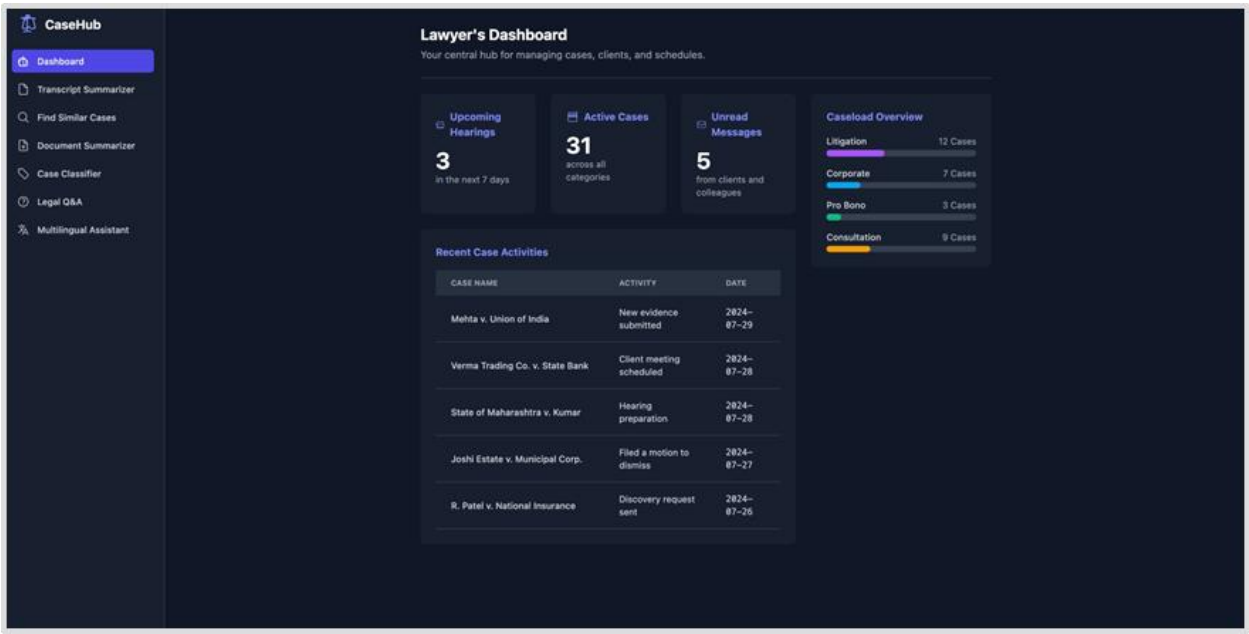

Fig 5.1.5 CASE HUB LAWYER DASBOARD

5.2 RESULT

The implementation of CaseHub proved highly effective in improving judicial workflows and communication. The system successfully enabled secure authentication, real-time case synchronization, and AI-based hearing summarization. Judges could view automatically generated summaries and similar past cases, while lawyers efficiently managed case lists and document uploads. The use of RAG and LLMs ensured factual, context-aware insights, and the integration of multilingual STT/TTS enhanced accessibility. Overall, CaseHub demonstrated increased efficiency, reduced manual workload, and provided reliable AI-assisted support, validating its role as an innovative AI-powered smart court management platform.

CONCLUSION

CaseHub successfully addresses the major challenges faced by the modern legal system, such as inefficient case management, poor communication among stakeholders, and fragmented workflows. By adopting a user-centered design thinking approach, the development process deeply analyzed the real frustrations and needs of judges, lawyers, clients, and administrative staff. These insights helped clearly define the core issues and guided the creation of a solution that is both practical and tailored to the complex ecosystem of the legal process.The platform unites judges, lawyers, clients, and court administrators on a single, integrated system that promotes seamless communication, real-time updates, and efficient management of cases and schedules. Through role-based access control, each user interacts with tools and features relevant to their responsibilities, enhancing productivity while maintaining data security.Features such as live hearing summaries, secure document sharing, voice note integration, and automated notifications work together to streamline legal proceedings and improve transparency. The modular design, coupled with modern technologies such as Firebase for real-time data syncing, AI for hearing summarization and retrieval, and robust security measures, ensures that CaseHub remains scalable, reliable, and user-friendly.Iterative testing and feedback from target users helped refine the system, making it responsive, accessible, and easy to navigate across multiple devices. Overall, CaseHub demonstrates how the thoughtful fusion of design thinking and advanced technology can revolutionize the legal sector by transforming slow, paper-based workflows into an efficient, collaborative, and transparent digital ecosystem.This project highlights the potential of AI-assisted smart courts, where automation and intelligence enhance judicial productivity, foster better collaboration, and accelerate the delivery of justice.

FUTURE WORKS

While CaseHub has successfully addressed core challenges in legal case management and communication, several enhancements are planned to further expand its capabilities and impact:

Planned Enhancements:

•Mobile App Development: Creating dedicated Android and iOS applications for smoother user experience and push notifications beyond the web interface.

•AI-Powered Case Insights: Integrating predictive analytics to estimate case outcomes, hearing durations, and optimal scheduling patterns.

•Voice Assistant Integration: Enabling hands-free voice commands and automatic transcription for quick note-taking and case updates.

•Offline Access: Introducing limited offline functionality to allow users to access essential case information without an internet connection.

•Feedback System: Adding real-time feedback and issue-reporting options for continuous improvement.

•Emergency Communication: Implementing quick alert features for urgent courtroom updates or legal emergencies.

Scalability Considerations:

CaseHub's modular and cloud-based architecture ensures scalability and adaptability across different court systems, jurisdictions, and institutional requirements. The flexible design allows easy customization and future expansion as user needs and legal technologies evolve.

As judicial systems continue to modernize, CaseHub will keep advancing with emerging AI, NLP, and automation technologies to remain a comprehensive, reliable, and user-centric platform. The ongoing goal is to make justice smarter, faster, and more collaborative through continuous innovation in AI-powered case handling and real-time judicial interaction.

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